

ID interesting case conference

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25th December 2024



Outlines

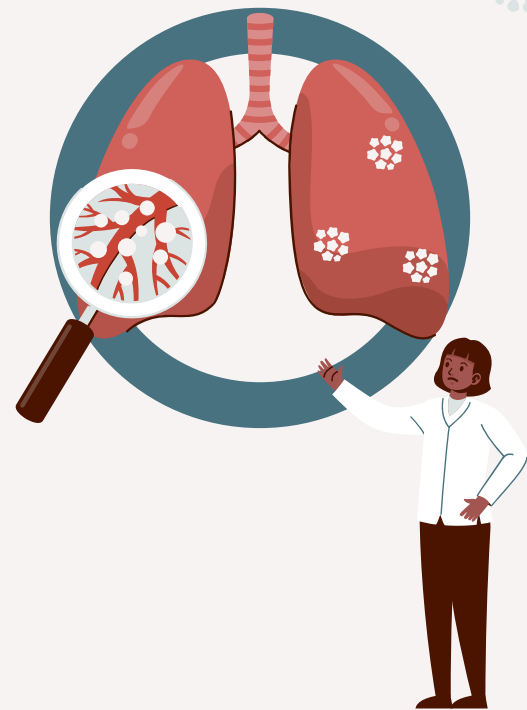
01 Case study

02 Review TB pericarditis (TBP) in children



Identification Data

- ผู้ป่วยเด็กชาย ชาวพม่า อายุ 11 ปี
- สิทธิการรักษา: ไม่ทราบสิทธิ
- แหล่งข้อมูล: ใบส่งตัว และญาติชาวพม่าช่วยแปลจากมารดา (พูดไทยได้เล็กน้อย)
- ความน่าเชื่อถือ: ต่ำ
- Admission date: 3/12/2567



Chief Complaint

- ตัวบวม 5 เดือนก่อนมารพ.



Present illness

- **1 yr PTA** ประมาณช่วงเข้าพรรษาปีก่อน (2/8/66) ที่บ้านพม่า **หน้าบวม ท้องบวม ขาสองข้างบวม** เวลาบวมจะบวมพร้อมกันทั้งหมด ไปพบแพทย์ที่รพ.พม่าแห่งแรก แจ้งว่า**ตับใหญ่** ให้ยาฉีดไม่ทราบชนิด (ไม่ได้ปัสสาวะออกมากขึ้นหลังฉีด) ได้นอนรพ. 4 คืน ได้ X-ray ตอนนอนรพ.อาการบวมลดลงทั้งหมด ไม่บวมแล้ว หลังจากนั้นออกรพ. หลังจากนั้น อาการบวมหายไปนาน 1 ปี ไม่เป็นๆ หายๆ เลย
- **5 month PTA** เริ่มกลับมามี**อาการบวมที่ท้องก่อน ค่อยๆ ขึ้นมาที่ใบหน้า และขาสองข้าง** แขนไม่บวม ขาบวมจนนั่งไม่ได้ จะเป็นหลังตื่นนอน ระหว่างวันหายไป บางวันเป็น บางวันไม่เป็น ไม่มีปัสสาวะเป็นฟองเลย ไม่มีปัสสาวะเป็นตะกอนปน ตอนนอนไม่รู้สึกลายใจเหนื่อย เวลาทานเผ็ดจะรู้สึกปวดท้อง ไม่อึดท้องเลย
- ไม่มีไข้ ไม่มีน้ำมูกนำมาก่อน ไม่ไอ ไม่ได้รู้สึกง่วงตัวเหลือง/ตาเหลืองมากขึ้น เหลืองมาตั้งแต่เกิดแล้ว
- ปฏิเสธประวัติ ไข้ขึ้นสูง ปวดตามข้อ ผื่นผดผื่นขึ้นตามใบหน้าหรือตัว ปฏิเสธประวัติคนในพื้นที่ ที่อยู่อาศัย หรือคนในบ้านไอนานเกิน 2 สัปดาห์

Laboratory work up at Lamphun hospital (13/12/67)

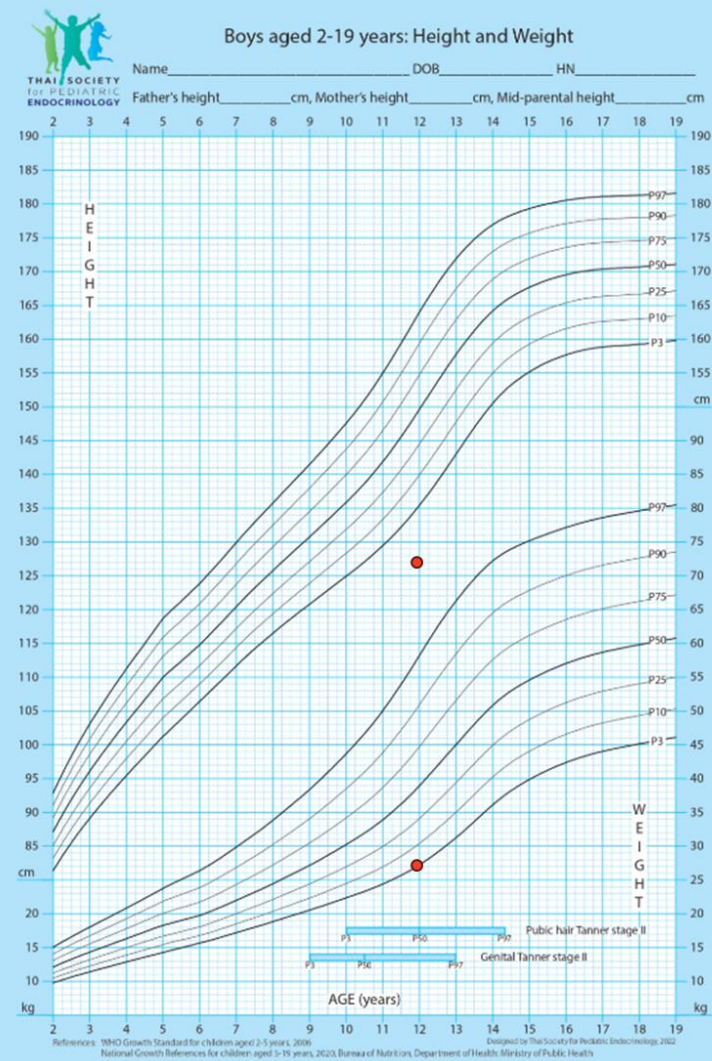
- CBC: Hb 10.3 g/dL, Hct 33%, WBC 7,210 cells/mm³ (N71.3%, L17.2%) platelet 282,000 cells/mm³, MCV 67.5 fL
 - BUN 10 mg/dL, Cr 0.49 mg/dL
 - LFT: TP 5.7 g/dL, albumin 3.0 g/dL, globulin 2.7 g/dL, AST 34 U/L, ALT 6 U/L, ALP 187 U/L, TB 0.4 mg/dL, DB 0.2 mg/dL
 - PT 13.0, PTT 29.8, INR 1.16
 - HBsAg quantitative <0.05: neg
 - Anti-HBs: <2.0 : neg
 - Anti-HCV: neg
- ☐ Refer CMU work up for chronic liver disease

Past history

- No known underlying disease
- Vaccine: รับที่พม่า แจ้งว่าได้ครบ ไม่ได้นำสมุดมา
- Natal history: คลอดเองที่บ้าน หลังคลอดแจ้งว่าปกติดี ไม่ได้ไปที่ รพ หลังเกิด ได้ไปฉีด vaccine ที่ รพ ใน อายุ 45 วัน
- ปฏิเสธประวัติสัมผัสผู้ป่วยวัณโรค

Growth curve

- BW 27 kg (P3)
- Ht 127 cm (<P3)
- W/A 71.05 %
- H/A 85.5 %
- W/H 108 %



Physical examination

- V/S: BT 36.5 °C, BP 101/70 mmHg, PR 92 bpm, RR 20/min, SpO2 100% (RA)
- BW 27 kg, Ht 127 cm
- **HEENT:** mild pale conjunctivae, anicteric sclera, no cervical LN can be palpated, jugular venous distension
- **Heart:** normal S1S2, no murmur, regular rhythm, jugular venous distension
- **Lungs:** no retraction, no chest wall deformity, decreased breath sound both lower lungs, no adventitious sound
- **Abdomen:** no distension, seen superficial vein dilatation, no spider nevi, normoactive BS, no abdominal bruit, soft, palpable mass at epigastrium, rubbery consistency, smooth surface, circular-shape extended from subxiphoid, size 10 cm, unmoveable, not tender, liver span 10 cm, spleen not palpable
- **Extremities:** no palmar erythema, no pitting edema, no skin lesion, no arthritis
- **Neuro:** awake and aware, well cooperated
- **PR:** yellow feces, good sphincter tone

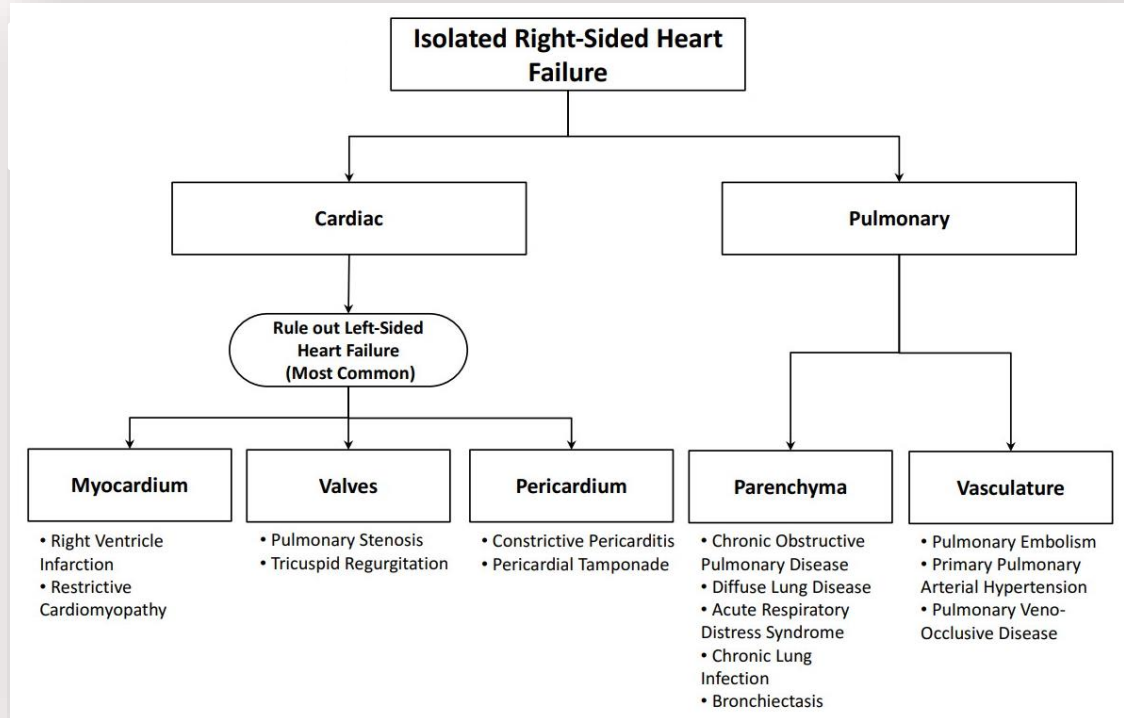




Problem lists

- Generalized edema with palpable epigastric mass
- Marked hepatomegaly with jugular venous distension

Causes of Right-sided heart failure



Investigations

CBC (3/12/67)

Hb	11.5 g/dL
Hct	37.2 %
WBC	8990 cells/cu.mm.
Neutrophil (%)	71.4 %
Eosinophil (%)	0.9 %
Basophil (%)	0.8 %
Lymphocyte (%)	18.0 %
Monocyte (%)	8.9 %
Platelet per cu.mm.	316000 cells/cu.mm.
MCV	65.6 fl. (80.0 - 99.0)
ANC	6420 cells/cu.mm.
ALC	1620 cells/cu.mm.
RDW	16.2 %

3/12/67

PT	13.00 sec (9.90 - 12.30)
INR	1.23 (>5.00)
PTT	36.70 sec (28.80 - 36.40) (>110.00)
PTT Ratio	1.15



Repeat Coagulogram
Hemato med

3/12/67

PT	12.7	sec (10.2-13.0)
INR	1.12	
PTT	28.1	sec (20.5-31.0)

Investigations

Chemistry (3/12/67)	Results
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BUN	12 mg/dl
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Creatinine	0.57 mg/dl
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Na	135 mmol/L
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K	3.0 mmol/L
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Cl	91 mmol/L
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Total CO2	32 mmol/L
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LFT (3/12/67)	Results
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Total Protein	6.5 g/dl
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Albumin	3.3 g/dl
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Globulin(cal)	3.2 g/dl
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Alkaline Phosphatase	217 U/L
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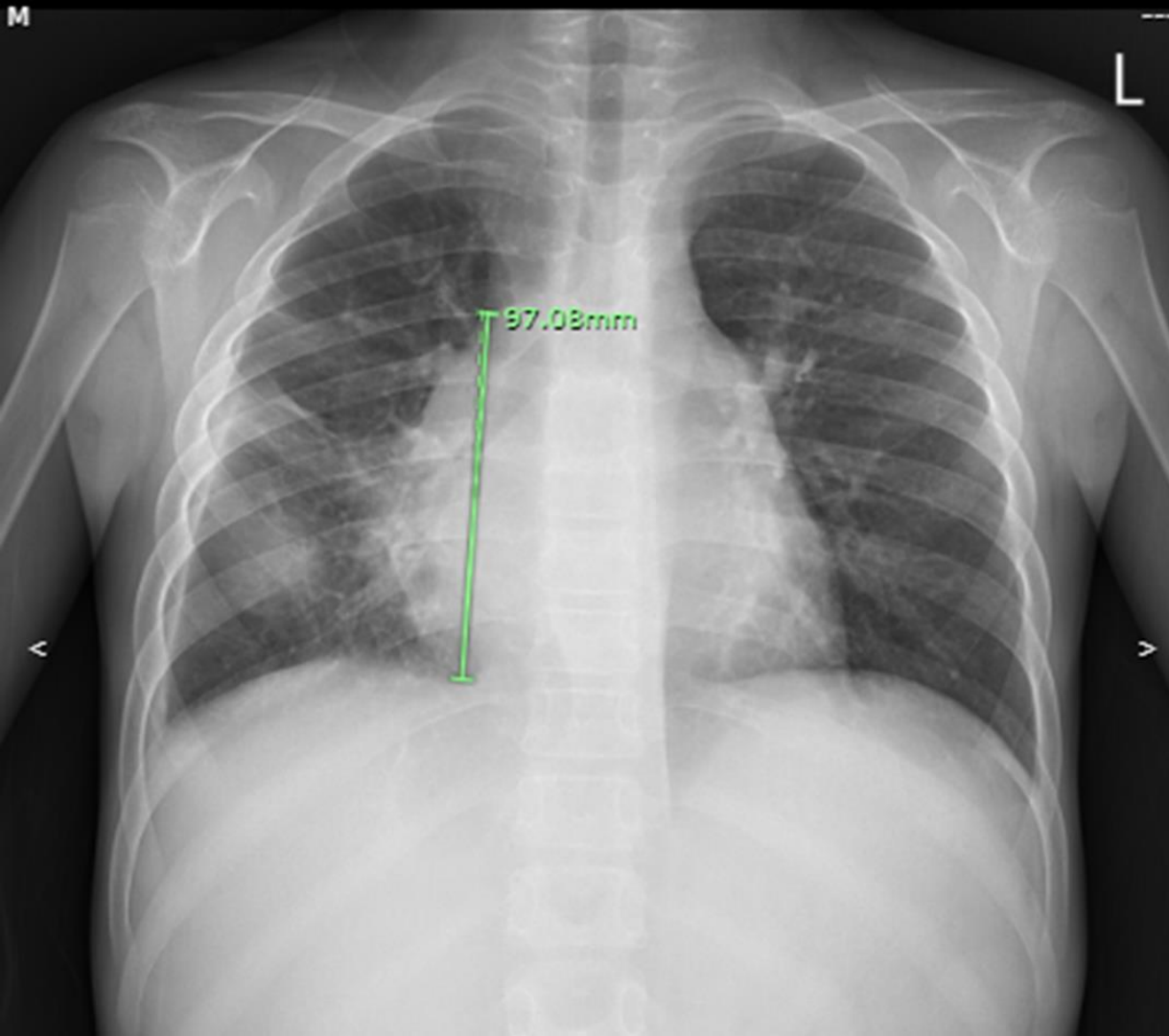
Cholesterol	257 mg/dl
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AST (GOT)	54 U/L
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ALT (GPT)	10 U/L
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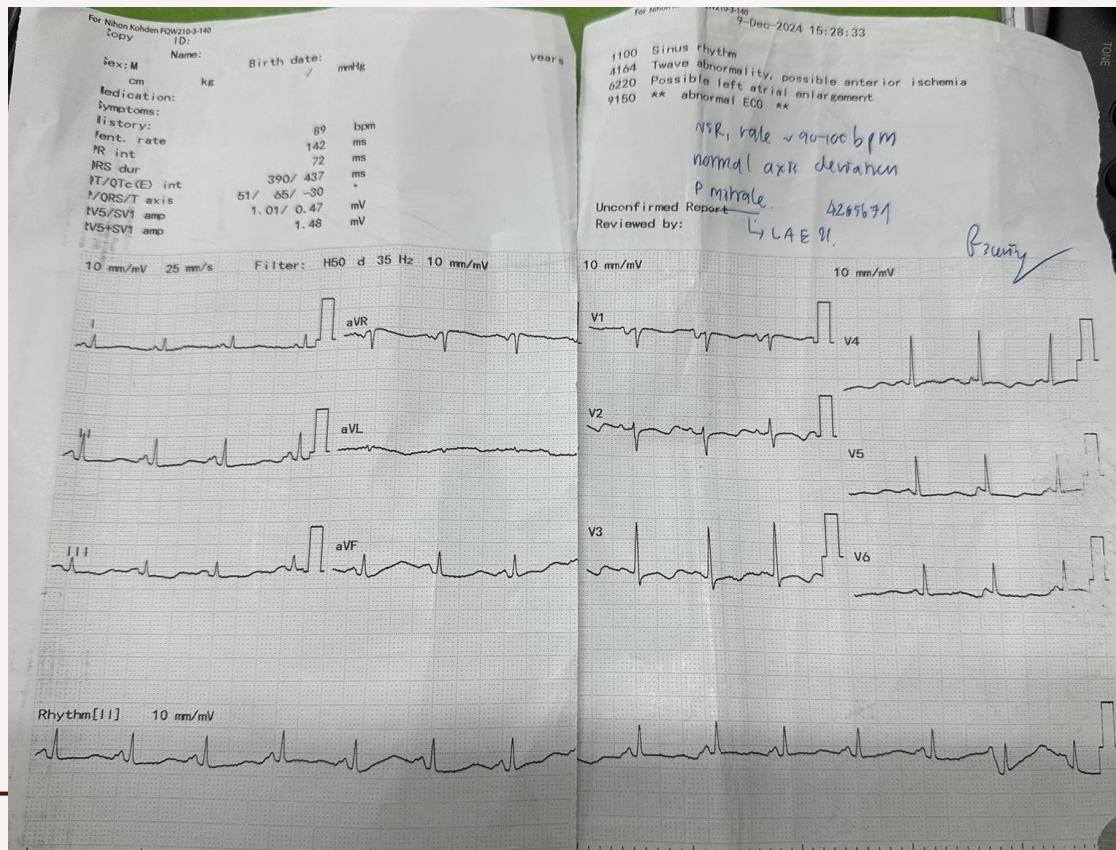
Total Bilirubin	0.39 mg/dl
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Direct Bilirubin	0.22 mg/dl
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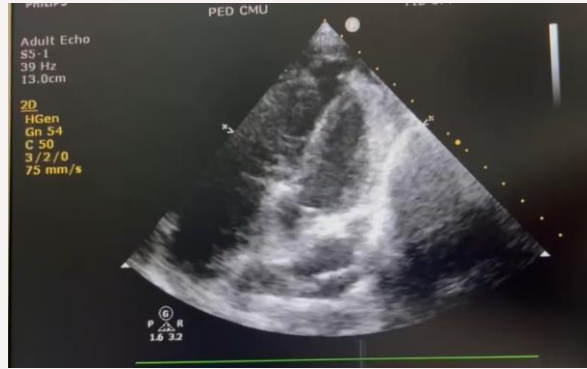
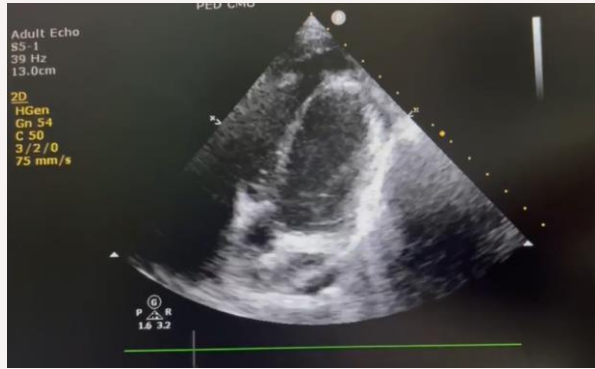


**CXR PA upright
(CMU)
3/12/2567**

EKG 12 leads



Echocardiogram



Consult cardio ped (11/12/67)

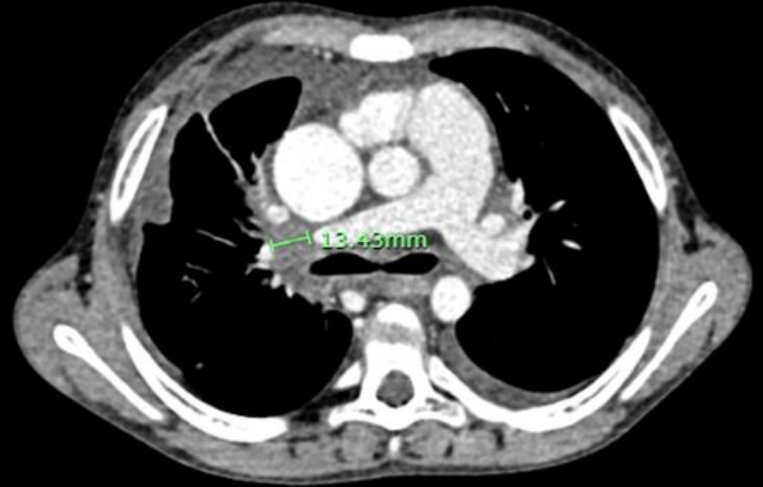
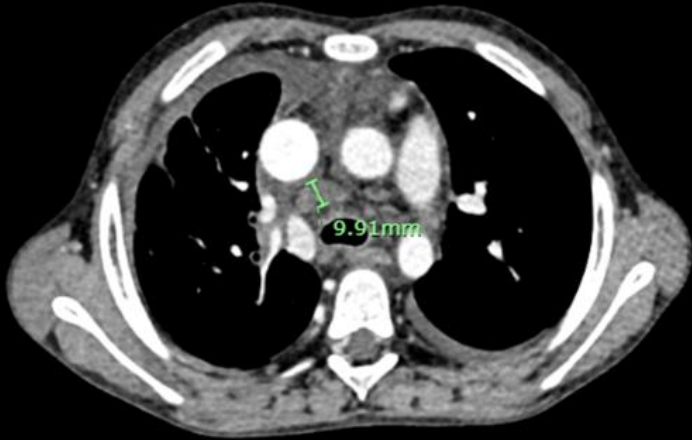
→ Echo bedside

- Constrictive pericarditis
- Good LVEF by eyeball

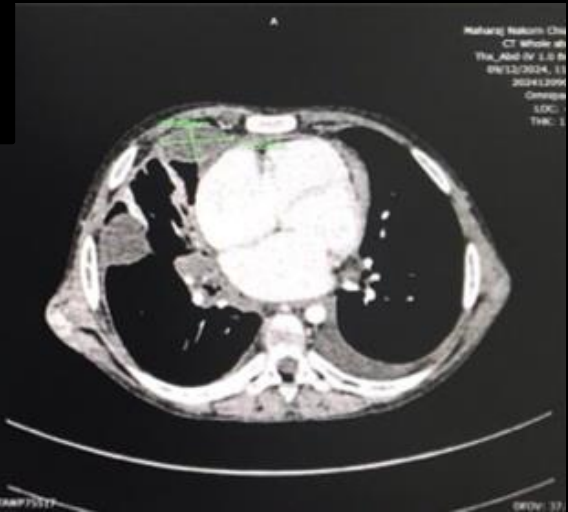
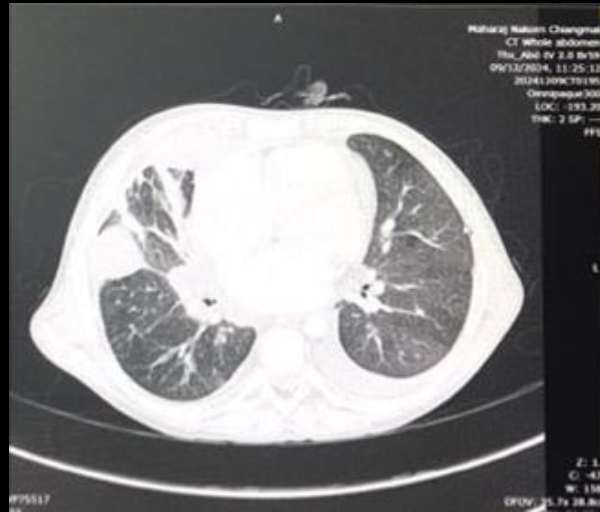
Urinalysis

Urinalysis		7/12/67
Color	Yellow	
Appearance	Clear	
pH	6.5 (4.6 - 8.0)	
Specific gravity	1.019 (1.003 - 1.030)	
Albumin	Trace	
Sugar	Negative	
Blood	1+	
Ketone	Negative	
Bilirubin	Negative	
Urobilinogen	2+	
Nitrite	Negative	
Leukocyte esterase	Negative	
RBC	1-2 HPF	
WBC	0-1 HPF	
Squamous Epithelium	None HPF	

CT Chest and abdomen (9/12/2567)



CT Chest and abdomen (9/12/2567)



CT Chest and abdomen (9/12/2567)



Chest

- Infiltrative soft tissue along right hilar and interlobar region extend to subcarinal region, size up to 1.3 cm. in axial diameter. This lesion could be lymphadenopathy.
 - **Multiple subcentimeter to centimeter lymph nodes** in the mediastinum, size up to 1.0 cm in short axis in the right lower paratracheal region.
 - **Multiloculated pleural effusions** along anterosuperior aspect of right hemithorax, size up to 2.0x3.9 cm and along right major fissure size 3.2x2.7 cm. No thickened enhancement of pleura or pleural nodule.
 - Small amount of left pleural effusion. No pneumothorax.
 - **Diffuse pericardial thickening with calcification**, measuring **0.5 cm in maximum thickness**. Noted bi-atrial enlargement, dilated SVC, IVC and hepatic veins.
- These findings are suggestive of **constrictive pericarditis**.
- A 1.5 cm focal pericardial defect at anterior aspect of right ventricle with herniation of RV size 2.3x1.8cm.
 - No worrisome pulmonary nodule or mass.
 - Multiple subsegmental atelectasis in right lung.
 - Patent trachea and bilateral main bronchi.
 - Unremarkable esophagus.
 - Normal enhancement of thyroid gland.
 - Intact bony thorax.

Abdomen

- Heterogeneous enhancement of the liver parenchyma with dilated IVC and hepatic veins, **likely from hepatic congestion**. No focal lesion. No dilatation of bile ducts. Patent hepatic and portal veins.
- Unremarkable gallbladder, pancreas, spleen, bilateral adrenal glands and bilateral kidneys.
- Unremarkable stomach and the visualized bowel loops.
- **Multiple subcentimeter lymph nodes** along gastrohepatic, perigastric, porta hepatis and mesenteric region, size up to 0.9 cm at mesenteric region.
- **Moderate amount of ascites** diffusely in abdomen. No peritoneal thickening or nodule.
- No pneumoperitoneum.
- Intact bony structures.
- Diffuse subcutaneous fat and mesenteric fat reticulation, could be systemic condition.

IMPRESSION:

- Suggestive of **constrictive pericarditis, TB pericarditis can not be excluded**.
- A 0.8 cm focal pericardial defect with herniation of RV. Please correlate with echocardiography.
- Infiltrative soft tissue along right hilar and interlobar extend to subcarinal region, could be lymphadenopathy including from TB.
- Multiloculated pleural effusions along right hemithorax and right major fissure.
- Hepatic congestion with moderate amount of ascites.



Impression

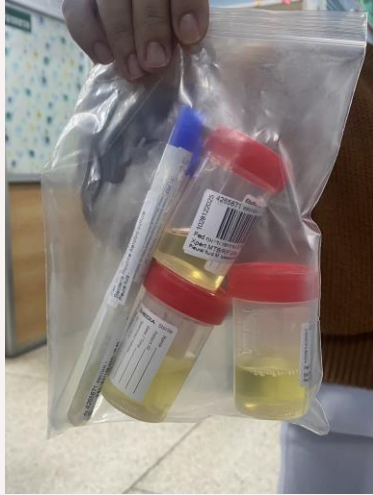
- Constrictive Pericarditis with suspected lymphadenopathy Ddx TB Pericarditis, malignancy
- Multiloculated pleural effusions along right hemithorax and right major fissure
- Hepatic congestion with ascites



Management

- W/U gastric washing for AFB stain, Xpert MTB/RIF ultra, mycobacterium C/S
- W/U stool for Xpert MTB/RIF ultra
- TST: 0 mm at 48 hr and 72 hr
- IGRA (Quantiferon TB) (13/12/67): negative
- Consult intervention for Tissue biopsy or pleural fluid aspiration

Operation: S/P Aspirate loculated Lt. pleural fluid (16/12/67)



Pleural fluid profiles	Findings
WBC	117 cells/cu.mm
RBC	30 cells/cu.mm
PMN	27%
MN	74%
Total protein	1.4 g/dL
LDH	41 U/L
PF TP/ serum protein ration	$1.4/5.7 = 0.24$
PF LDH/ serum LDH	$41/232 = 0.18$

Pleural fluid work up

Pleural fluid profiles	Findings
AFB stain	No AFB found
Xpert MTB/RIF ultra	MTB not detected
Mycobacterium C/S	pending
ADA	11 U/L

LABORATORY REPORT

Patient Name

Address/Ref.No. : General Patient (กุมารเวชศาสตร์ 3 ชั้น 6 ตึกบ
ญสม)/HN.4265671

Sex : Male DOB : --- Age : 11Y

MRN. : NCLA-24-023508

Requested Date : 19 Dec 2024 15:32

Lab No. : NCL243540087

Received Date/Time : 19 Dec 2024 15:32

Collected Date/Time : 19 Dec 2024 15:31



Doctor : หมอระบุนแพทย์

Test Name	Result	Unit	Reference Range
Adenosine Deaminase			
Specimen.....:	Pleural Fluid		
Adenosine Deaminase(Pleural Fluid):	11.0	U/L	(0.0-33.0)
Method.....:	Colorimetric		



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2301/2 New Petchburi Road, Soi 47 (Soonvijai)
Bangkapi, Huaykwang, Bangkok 10310, Thailand
Tel.(662) 762-4000 Fax (662) 762-4196
Email: nbcslab@nhealth-asia.com



Accreditation No. 4120/47
15189:2022

LABORATORY REPORT

Patient Name

Address/Ref.No. : General Patient (ผู้ป่วย 3 ชั้น 6 คือกลุ่มสม)/HIN.
4265671

Sex : Male DOB : --- Age : 11Y

Requested Date : 13 Dec 2024 17:43

Received Date/Time : 13 Dec 2024 17:43



MRN. : NCLA-24-023508

Lab No. : NCL243480139

Collected Date/Time : 13 Dec 2024 17:42

Doctor : นวรัตน์แพทย์

Test Name	Result	Unit	Reference Range
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(*) Quantiferon TB

Specimen.....: Plasma
Interferon gamma TB1.....: 0.010 IU/mL
Interferon gamma TB2.....: 0.010 IU/mL
Quantiferon TB.....: Negative
Interpretation.....: M.Tuberculosis infection not likely
Remark : If one or both of the QFT-Plus TB antigen tubes are equal to or
greater than 0.35 IU/ml, and greater than 25% of the Nil tube value,
the patient is considered positive

Remark:

- 1: Diagnoses of M. tuberculosis infection and decisions about medical or public health management should not be based on IGRA or TST results alone, but should include consideration of epidemiologic and medical history as well as other clinical information.
- 2: Repeating an IGRA or performing a TST might be useful when the initial IGRA result is indeterminate, borderline, or invalid and a reason for testing persists.

Pleural effusion cytology

CYTOLOGICAL DIAGNOSIS:

Pleural effusion, right, thoracentesis:

- Negative for malignancy

MACROSCOPIC DESCRIPTION:

Received is 4.5 ml of clear yellow fluid. The specimen is insufficient for cell block preparation

MICROSCOPIC DESCRIPTION:

Two megachamber preparation slides are examined. Mesothelial cells are scattered among leukocytes, mostly small lymphocytes.



Investigations

- Anti-HIV: non-reactive
- **Hepatitis profiles**
 - HBsAg: negative
 - Anti-HBs: negative 4.74 mIU/ml
 - Anti-HCV: negative



Management

- Start anti-TB drugs → Isoniazid, rifampicin, pyrazinamide, ethambutol
- Monitor side effect: N/V, abdominal pain, hepatitis
- Directly-Observed Treatment (DOT)
- F/U CT chest next 2 months after start anti-TB drugs



Other problems

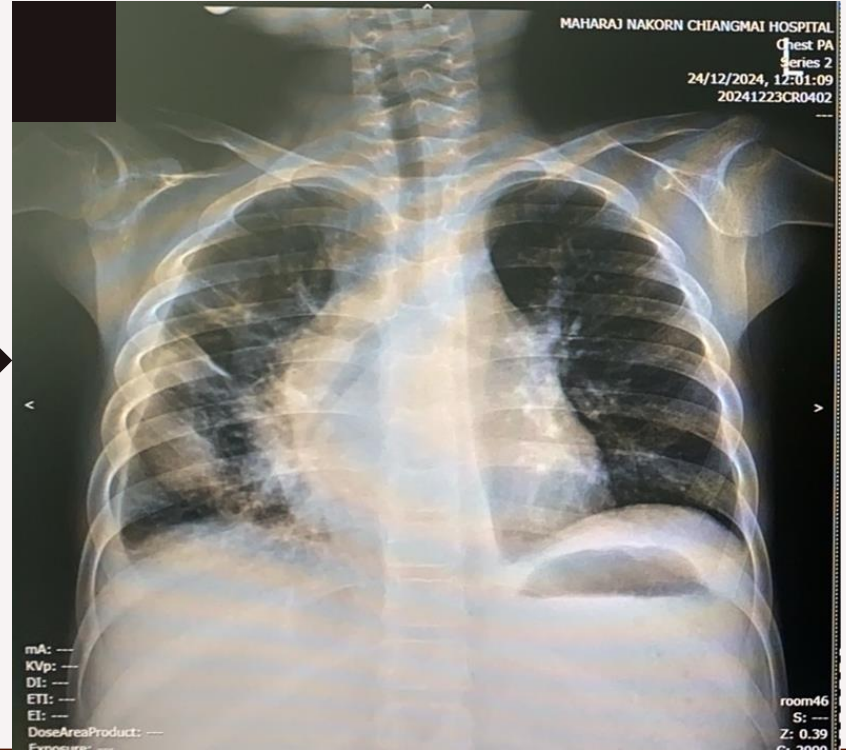
Suspected protein losing enteropathy from secondary lymphangiectasia (TB)

Consult GI:

Plan:

- Stool Alpha-1 antitrypsin
- Add high protein diet

CXR Progression 1 week after start anti-TB drugs





TB pericarditis (TBP) in children

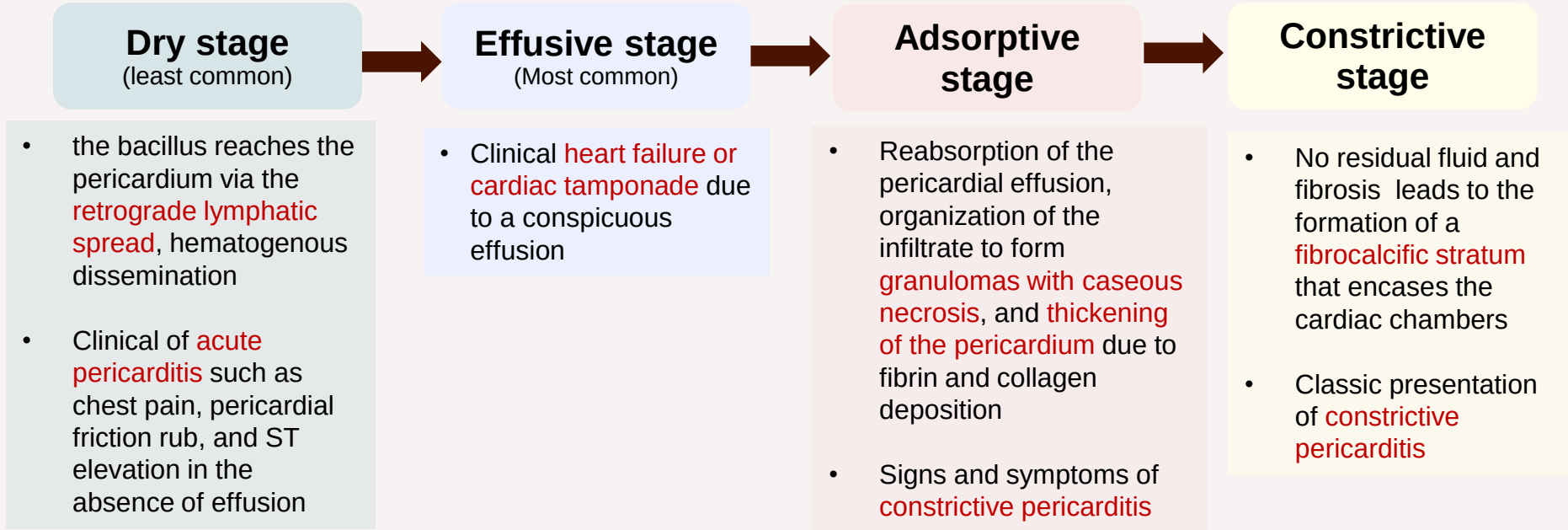


Introduction

- Tuberculous pericarditis (TBP) is a rarely seen form of extrapulmonary TB, accounting for **1–2% of all TB infections**
- TBP is the most common cause of massive pericardial effusion in developing countries.
- It is the **most common cause of pericarditis in Africa and other countries** in which TB remains a major public health problem

Progression of Tuberculous Pericarditis

- 4 stages



Clinical manifestations of TB pericarditis in children

- Fever
- Cough
- Weight loss
- Right-sided heart failure
- Hepatomegaly
- Jugular vein turgor
- Others (infrequent)
 - Chest pain, friction rub, and muffled heart sounds
 - Cardiac tamponade

Relevant Reported Signs and Symptoms	Number of Patients	Frequency
fever	76	62%
hepatomegaly	74	60%
cough	72	59%
dyspnea	64	52%
weight loss or low body weight	60	49%
increased JVP or JV turgor	46	37%
chest pain	28	23%
lymphadenopathy	24	20%
tachypnea	23	19%
edema	22	18%
pulsus paradoxus	21	17%
tachycardia	17	14%
abdominal distention	16	13%
ascites	15	12%
muffled heart sounds	13	11%
friction rub	13	11%
abdominal pain	12	10%
splenomegaly	10	8%
pulmonary edema or orthopnea	10	8%
vomiting	9	7%
night sweats	8	7%
fatigue	9	7%
gallop rhythm	4	3%



Clinical presentation of TB pericarditis

- 3 forms

1

Pericardial effusion

2

Constrictive pericarditis

3

Combination of effusion and constriction

Constrictive pericarditis

- One of the most serious sequelae of tuberculous pericarditis, occurring in **30% to 60%** of patients, despite prompt antituberculosis treatment
- **Clinical presentation:**
 - Highly variable, ranging from **asymptomatic to severe constriction**
 - the diagnosis is often missed on cursory clinical examination
- **Physical examination:**
 - Diastolic lift (pericardial knock) with a high-pitched early diastolic sound
 - Sudden inspiratory splitting of the second heart sound are subtle but specific physical signs



Diagnosis of TBP

- Most patients with constrictive pericarditis have the **subacute variety**
 - thick fibrinous exudate fills the pericardial sac, compressing the heart and causing a circulatory disturbance
- **Chest radiographic findings:**
 - Massive pericardial effusion (globular enlargement)
 - Constrictive
 - Nonspecific
 - Absence of the notch at the root of the right lung and a distended superior vena cava



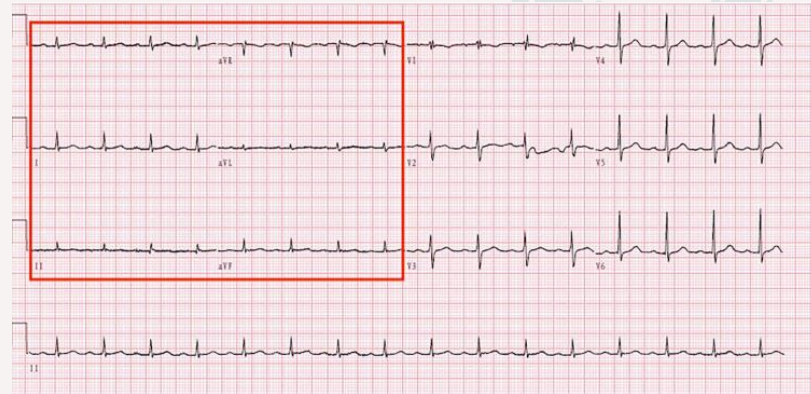
Diagnosis of TBP

- **Electrocardiogram:**

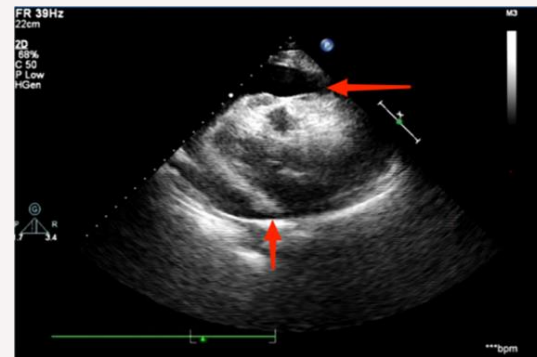
- Most cases nonspecific but generalized T-wave changes
- **Low-voltage complexes** (30% of cases)

- **Echocardiogram:**

- Pericardial effusion
- Constrictive pericarditis
 - Thick fibrinous exudate in the pericardial sac
 - Diminished movements of the surface of the heart
 - Normal-size chambers, absence of valvular heart disease, and absence of myocardial hypertrophy



EKG: low voltage in the limb leads



Echocardiography suggests the presence of a moderate to large pericardial effusion

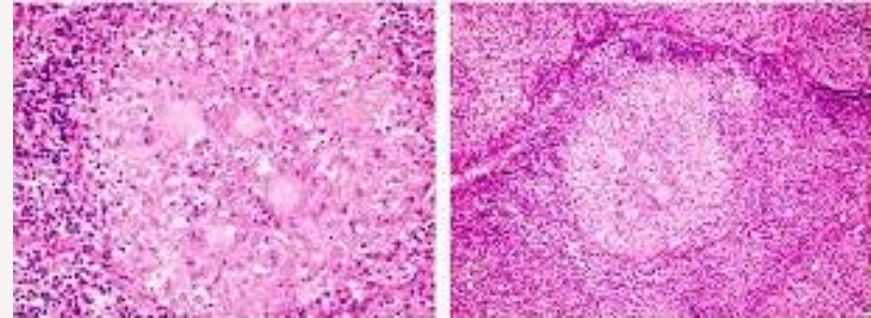


Laboratory tests for diagnosis TBP

- **Xpert MTB/RIF**
 - Sensitivity 68 % and specificity 99%
- **ADA**
 - ≥ 40 U/L usual when diagnosing the possible presence of TBP
 - Sensitivity 84%, specificity 80.0%
- **Pericardial fluid culture**
 - Not preferred due to time-consuming

Histological diagnosis

- Biopsy samples are infrequently collected because of a lack of diagnostic suspicion or due to the invasive nature of the procedure
- **Findings:**
 - granulomas, chronic inflammatory infiltrates with histiocytes, and fibrosis)
- Presence of MTB in the pericardium (fluid or tissue)





Imaging

- **CT scan:**
 - Demonstrate pericardial effusion, pericardial thickening and enhancement.
 - Organ involvement e.g. pulmonary tuberculosis, mediastinal lymph node involvement
- **MRI scan:** adjunctive imaging modality for pericarditis
 - Pericardial thickening
 - Pericardial edema
 - Heterogeneous pericardial effusion
 - Pericardial enhancement



Sites affected by TB other than the pericardium

Other TB Manifestations	Frequency
Pulmonary TB	42%
Pleural effusion	37%
Lymphadenopathy	30%
Disseminated or miliary TB	15%
Abdominal TB	14%
TB Meningitis	3%
Cardiac	3%
TB Abscesses	2%



Complications of tuberculous pericarditis

- Cardiac tamponade
- Pericardial constriction
- Hepatic congestion
- Cardiac arrest and death



Treatment of TBP

- **Regimen**
 - “quadruple” regimen of rifampin, isoniazid, pyrazinamide, and ethambutol for 2 months, followed by isoniazid and rifampin for 4 months
- **Adjuvant corticosteroid therapy**
 - reduces the hospitalization rate for TBP and the incidence of patients progressing to constrictive pericarditis
- **Pericardiocentesis**
 - Routine pericardiocentesis with prolonged drainage may reduces the incidence of constrictive pericarditis

Outcome

- TBP has an **excellent prognosis** in children when diagnosis is timely and treatment is available
- Poor prognosis and high mortality rate owing to constrictive pericarditis



Summary

- The diagnosis of TB pericarditis is a challenge
 - no definitive diagnostic evidence - mycobacterium tuberculosis
- Negative routine laboratory tests for the pericardial effusion of unknown origin does not exclude tuberculosis pericarditis.
- And a biopsy of the pericardium may help clarify the diagnosis based on excluding other causes.
- Empirical anti-TB treatment may be considered after non-TB causes have been ruled out to the extent possible.



Thank you